A Multivariate Look into Governance Quality Indicators

Daniel Cowley

March, 2026

1 Introduction

The goal of governments should be to establish the rule of law, carry it out, and serve its people. Tracking the effectiveness of governments over time is a good way to gauge the success of these goals globally. In this study, we examine the World Governance Indicators (WGI) report from the World Bank (Kaufmann et al., 2010), which aggregates expert and citizen assessments of governance quality across 214 countries from 1996 to 2023, making for 5045 observations. Using Factor Analysis, we aim to address the following questions: Does governance have a multi-dimensional construct? How do geographical regions differ on these dimensions, and have these regional differences shifted over time?

2 Data

2.1 Variables

There are six aggregated indicators in the dataset with the following brief summaries (full descriptions may be found in Kaufmann et al. (2010)): Control of Corruption measures the extent to which public power is exercised for private gain. Government Effectiveness captures the ability to formulate quality policies, implement them, and the government's commitment to carry them out. Political Stability reflects the likelihood of politically motivated violence. Regulatory Quality assesses the government's ability to develop and implement healthy policies that promote private sector development. Rule of Law measures the confidence and adherence to societal rules and protection from crime. Voice and Accountability captures the degree to which citizens can participate in selecting their government, freedoms of expression, association, and the press.

Scores in these categories have been centered and scaled, resulting in a range of approximately -2.5 (weakest governance) and +2.5 (strongest governance). Thus, units represent the standard deviations from the cross-country mean.

2.2 Regional Classification

Country data was grouped by the following nine geographical regions to explore regional differences in governance quality: Sub-Saharan Africa (n = 50), East Asia & Pacific (n = 37), Eastern Europe & Central Asia (n = 29), Central America & Caribbean (n = 27), Western Europe (n = 25), Middle East & North Africa (n = 20), South America (n = 13), South Asia (n = 8), and North America (n = 5). It is important to note that regional assignments are subjective and may be debated. One example is Greenland's assignment to North America while it is a territory of the Western European country Denmark. A map of country-region assignments may be found in Figure 6.

2.3 Temporal Groupings

We used two temporal groupings of the data in our analyses, the first being the mean, and the second was through the construction and comparison of two "time windows." These time windows are defined as early (1996-2000) and recent (2019-2023). Each window was chosen to span 5 years and average the score over the window to account for missing data and to smooth the effects of short-term major events (e.g. the September 11th attacks in the United States).

3 Exploratory Data Analysis

3.1 Data Inspection

Taking a look at the data itself, from year to year, some countries are occasionally absent from the data. This is most notable in the early years of data collection, where from 1996 to 2003, approximately 30 countries were missing. Most years however, only around 5 countries are absent, likely due to factors such as the dissolution of states, regime changes, or other geopolitical circumstances. One notable case is the small island nation of Niue, located in the South Pacific Ocean, which has recorded data points only for 2009, 2010, and 2022, which will appear again in Discussion (Section 7). As scores were already standardized within the dataset, these missing data outliers were retained as they were not expected to negatively impact subsequent analyses.

3.2 Distributional Characteristics

Figure 1 presents the distributional characteristics across regions using three metrics: mean governance score by indicator, composite score across all observation years (1996-2023), and composite score compared across early (1996-2000) and recent (2019-2023) windows. The violin plots show strong bimodal distributions within North and South America, indicating that there may be interregional divides in governance indicators unaccounted for by the groupings. Western Europe shows the lowest variance among regions, likely reflecting the consistency of governance quality among members of the European Union.

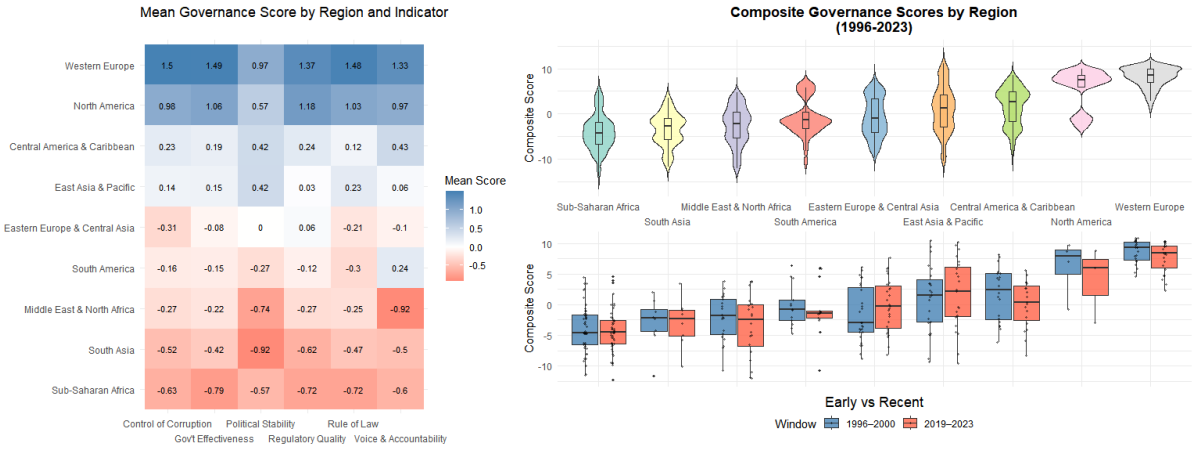


Figure 1: **Regional Distributional Characteristic Plots.** *Left:* Mean Governance Score by Indicator. Shows a clear difference in the mean across regions and strong correlation among governance indicators, with the exception of Political Stability. *Top Right:* Composite Governance Score Violin Plots. Bimodal distributions within North America and South America suggest meaningful within-region heterogeneity not captured by geographic groupings. *Bottom Right:* Early (1996-2000) and Recent (2019-2023) Time Windows Boxplot Comparison. Shows distinct shifts in regional governance over time, with North America and Western Europe declining, while Eastern Europe & Central Asia improved.

All three subplots motivate the stratification by region, as regional means differ substantially across indicators, time windows, and the composite score.

3.3 Regional Differences

The mean governance score heatmap in Figure 1 shows that while most indicators are strongly correlated across regions, Political Stability consistently stands out as the weakest, suggesting it captures a dimension of governance not fully shared with the remaining indicators. This observation is supported by the correlation heatmaps discussed in Section 4.1.

3.4 Temporal Trends

The early versus recent window boxplot comparison in Figure 1 shows distinct shifts in regional governance over time. Most notably, the two strongest governance regions, North America and Western Europe, have both declined, while Eastern Europe & Central Asia showed noticeable improvement. It is important to note that North America's decline may be exaggerated given the small number of countries in this region ($n = 5$).

4 Assumption Testing

4.1 Correlation Structure

Looking at the correlation structure of this dataset, the global Pearson correlation heatmap in Figure 2 shows strong positive correlations among all six governance indicators, ranging from $r = 0.67$ between Political Stability and Regulatory Quality to $r = 0.94$ between Control of Corruption and Rule of Law. This is expected, as all indicators reflect underlying governance quality. However, examining regional heatmaps separately reveals that some of these correlations are inflated by aggregating all regions. Most notably, Political Stability's correlation with Voice & Accountability drops to ($r = 0.11$) in the Middle East & North Africa, suggesting that civilian security and democratic accountability do not affect each other in regions that regularly experience conflict.

This pattern of high shared correlation globally, combined with meaningful regional variation, motivates the use of Factor Analysis (FA) to recover latent governance dimensions underlying the observed indicators. Our aim in subsequent sections will be to justify the choice of FA.

4.2 Multivariate Normality

To further support our selection of FA to model the data, we must confirm the multivariate normality assumption. Mardia's test is applied first, where skewness passes ($p \approx 1$), while kurtosis fails ($p \approx 0$). This failure is likely due to the fact that kurtosis is a fourth-order statistic, which is highly sensitive to both outliers and large sample sizes. Given that no extreme outliers are present in the data, as indicators are standardized to [-2.5, +2.5], the kurtosis rejection is likely an artifact of power due to the large sample size (n = 5045), rather than a meaningful departure from normality.

To corroborate this finding with a test more robust to large sample sizes, negentropy is computed for each indicator, where values range from 0.01 to 0.03 across all indicators, confirming Gaussianity as all values are near zero. These results are consistent with Mardia's skewness result.

The support of Mardia's skewness and negentropy provides sufficient evidence to proceed under the assumption of multivariate normality. This supports our choice of FA, which assumes Gaussian latent factors, in contrast to Independent Component Analysis (ICA), which relies on non-Gaussianity for identifying signal.

4.3 Sampling Adequacy

Lastly, we assess sampling adequacy across three criteria. First, with an n/p ratio of 840, the sample size far exceeds the threshold for stable solutions, ensuring convergence of the distribution. Second, the Kaiser-Meyer-Olkin (KMO) measure yields an overall KMO of 0.9, classified as "marvelous" under Kaiser's rule of thumb (Kaiser, 1974). This is consistent with the prior observation in the global Pearson correlation heatmap, where indicator correlations ranged from $r = 0.67$ to $r = 0.94$, indicating strong shared correlation. Finally, Bartlett's test of sphericity strongly rejects ($p \approx 0$) that the null correlation matrix is an identity matrix, confirming there is meaningful correlation structure (Bartlett, 1951). Having established that the data satisfy the necessary assumptions, and given that interpretability of latent governance constructs is the primary goal of this analysis rather than dimension reduction, FA is preferred over Principal Component Analysis.

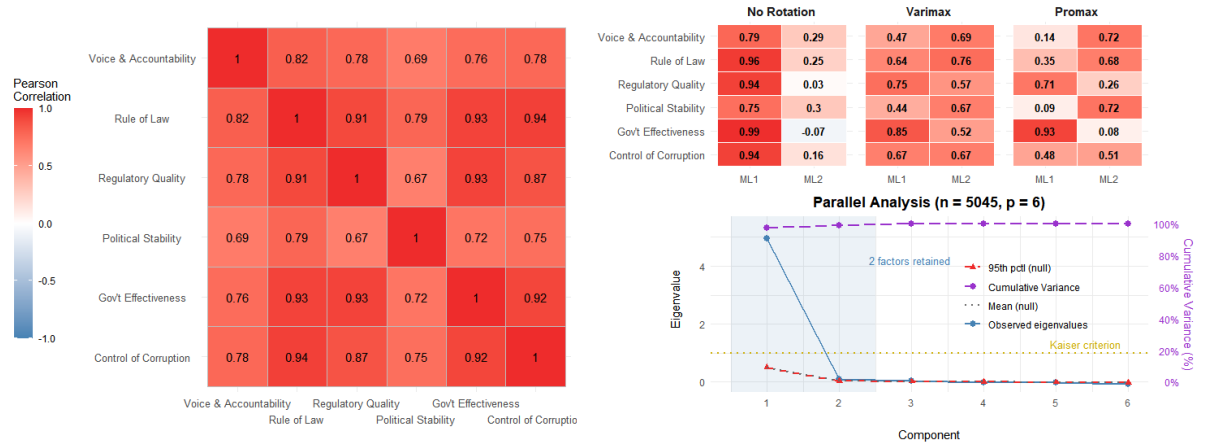


Figure 2: **Pearson Correlation, Factor Loadings, and Parallel Analysis Scree Plot.** *Left:* Global governance indicator Pearson correlation heatmap. Shows all governance indicators are strongly correlated, ranging from $r = 0.67$ to $r = 0.94$, giving early motivation to use FA. *Top Right:* Loading heatmaps under no rotation, Varimax, and Promax. Promax offers the strongest interpretability, with Factor 1 capturing State Effectiveness with high loadings on Government Effectiveness (0.93), Regulatory Quality (0.71), and Control of Corruption (0.48), and Factor 2 capturing Civil Order & Accountability through Voice & Accountability (0.72), Political Stability (0.72), Rule of Law (0.68), and Control of Corruption (0.51). *Bottom Right:* Scree plot with various criteria used in the model specification process for FA. Two factors were selected for interpretability.

5 Factor Analysis

5.1 Model Specification

We first justify the choice of Maximum Likelihood (ML) estimation over Principal Axis Factoring (PAF). While PAF is more robust to multivariate normality violations, ML is preferred in this case, as we established that multivariate normality holds in Section 4.2 through Mardia's skewness test ($p \approx 1$) and near-zero negentropy values (0.01 - 0.03).

To determine the number of factors to retain, we used parallel analysis alongside the scree plot in Figure 2, which shows various criteria. The scree plot exhibits a sharp elbow after the first factor, immediately suggesting that variance is dominated by a single factor. This is confirmed by the cumulative variance, where the first factor alone accounts for 97.41%, well above the conventional 80-90% threshold. The Kaiser criterion further supports retaining only one factor, with eigenvalues $\lambda_1 \approx 4.95$ and $\lambda_2 \approx 0.11$, yielding an approximate 48:1 ratio indicating high stability. Parallel analysis agrees, as only the first eigenvalue exceeds the 95th percentile of simulated null eigenvalues.

However, despite all criteria unanimously pointing to a one-factor model which may be interpreted as a general governance factor, we choose to retain a second factor for explanatory purposes. As observed in the regional correlation heatmaps and the mean governance score heatmap (Figure 1), Political Stability has notably weaker correlations with the remaining indicators, thus motivating a two-factor solution that may help capture this discrepancy that a single general governance factor cannot express. For this second factor to be meaningfully interpretable however, an appropriate rotation method must be selected.

5.2 Rotation Selection

Looking at the loading heatmaps for no rotation vs Varimax vs Promax in Figure 2, we notice that Promax rotation offers the strongest interpretability by producing two clearly distinguishable factors. Under Promax, Factor 1 has strong loadings of Government Effectiveness (0.93), Regulatory Quality (0.71), and Control of Corruption (0.48), collectively may be interpreted as reflecting the government's ability to formulate, implement, and enforce public policy without being undermined by private interests to corrupt. The second factor, with strong loadings of Voice & Accountability (0.72), Rule of Law (0.68), Political Stability (0.72), and Control of Corruption (0.51), reflects the degree to which citizens feel safe, represented, and protected under their government. Note that corruption loads onto both factors, which shows that corruption undermines both institutional effectiveness and citizen trust at the same time.

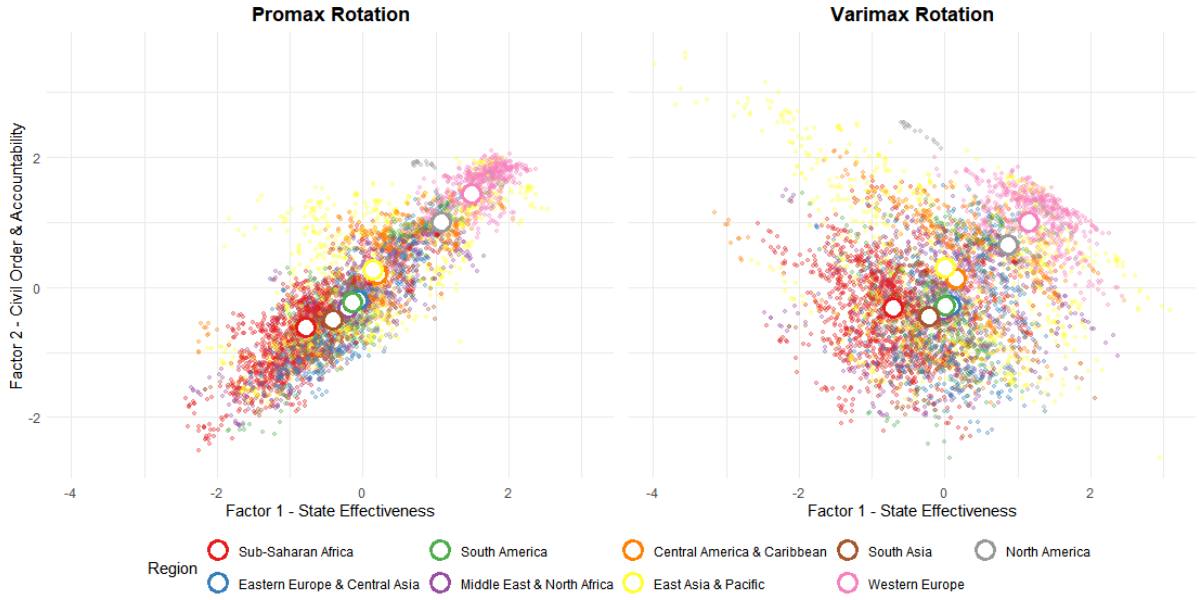


Figure 3: **Score Biplot Comparison of Varimax and Promax Rotation.** *Left* shows Promax rotation and *right* shows Varimax rotation. Both biplots are stratified by region with centroids displayed. Promax was selected as the preferred rotation, as the natural intercorrelation ($r \approx 0.85$) between State Effectiveness (Factor 1) and Civil Order & Accountability (Factor 2) is preferred over having orthogonality imposed by Varimax.

The factor score biplots in Figure 3 further illustrate the difference between rotation methods found in Figure 2. The Promax solution produces a diagonal distribution, reflecting the permitted intercorrelation between factors ($r \approx 0.85$). This strong correlation shows that countries that score high in the first factor's loadings: Government Effectiveness, Regulatory Quality, and Control of Corruption, likely do well in the other governance indicators too. By contrast, the varimax solution enforces orthogonality, working against the natural correlation of the two factors; as such, Promax rotation will be used in all subsequent analyses.

6 Interpretation

6.1 Regional Governance Profiles

The Global factor score biplot in Figure 3 displays distinct separation between regional centroids, supporting the decision to group the analyses by region. Western Europe occupies the highest position on both factors, while Sub-Saharan Africa the lowest.

Examining the regional trajectory biplots in Figure 4, the bimodal distributions noted in the composite score violin plots (Figure 1) are apparent again within North America, South America, and South Asia, suggesting there are more meaningful within-region differences that the geographic groupings used cannot fully capture.

To formally test whether regional governance profiles shifted significantly between early (1996-2000) and recent (2019-2023) windows, a bootstrap test ($B = 1000$) was conducted on the Euclidean distance between the regional centroid in the factor space.

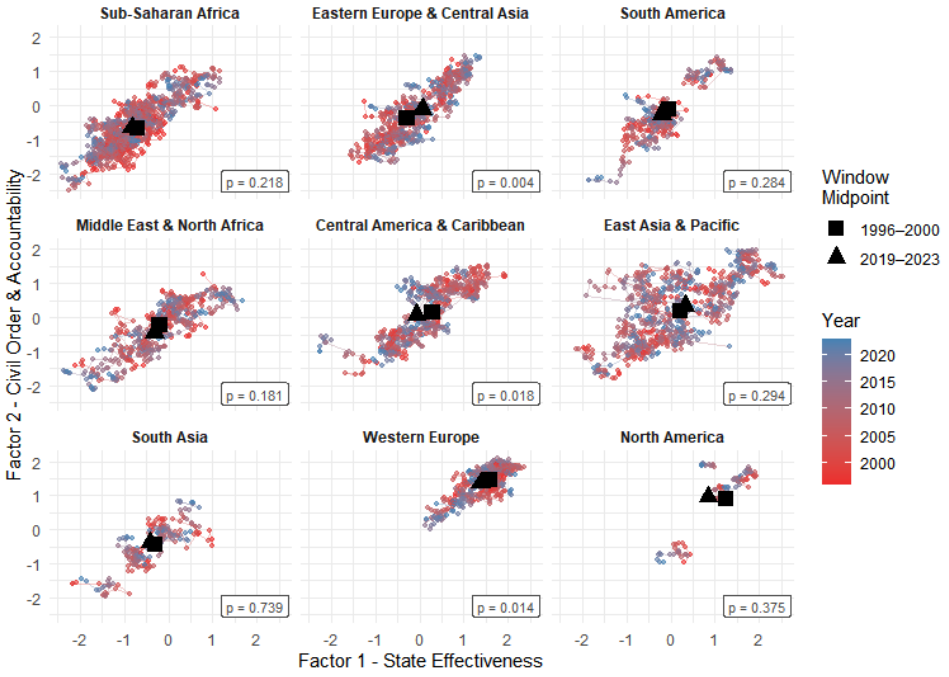


Figure 4: **Regional Governance Trajectories in Factor Space.** Each subplot shows the shift in regional centroids between the early (1996-2000) and recent (2019-2023) windows across State Effectiveness (Factor 1) and Civil Order & Accountability (Factor 2). p-values are derived from a bootstrap test ($B = 1000$) on the Euclidean distance between centroids. Eastern Europe & Central Asia showed the largest improvement ($p = 0.004$), while Central America & Caribbean ($p = 0.018$) and Western Europe ($p = 0.014$) both regressed on Factor 1.

The results from the bootstrap test show differing trends across regions. Eastern Europe & Central Asia showed the largest and most significant improvement ($p = 0.004$), advancing on both factors. By contrast, Central America ($p = 0.018$) & Caribbean and Western Europe ($p = 0.014$) both regressed on Factor 1. It's important to note that regions with smaller country counts, e.g. North America ($n = 5$), may be inaccurate as the bootstrap test has limited power at small sample sizes. Next, we will examine the individual countries influencing these regional trends.

6.2 Country-Level Analysis

Figure 5 presents two country-level score profiles of interest, examining governance change over time and imbalances between the two factors.

The left subplot shows the nine countries with the largest Euclidean trajectory distance in factor space between the early (1996-2000) and recent (2019-2023) windows. The countries with the largest improvement are Rwanda (dist. = 2.04), Georgia (dist. = 1.84), and Serbia (dist. = 1.46), all of which began the observation period among the lowest-scoring countries and progressed to average governance profiles. This trajectory is consistent with positive regime changes. In contrast, the countries that regressed most severely are Venezuela (dist. = 1.82), Syria (dist. = 1.79), Yemen (dist. = 1.63), and Lebanon (dist. = 1.34), reflecting the institutional collapse associated with prolonged conflict, civil war, and state failure.

The right subplot identifies countries whose average factor scores deviate furthest from the Factor 1 = Factor 2 diagonal. Given the strong intercorrelation ($r \approx 0.85$) found previously, most countries cluster near this diagonal, making such deviations interesting cases. The countries furthest from the diagonal are Tuvalu (dist. = 1.3), the Marshall Islands (dist. = 1.18), and Kiribati (dist. = 0.9). A notable commonality between these countries is that they are all small island nations, several of which are territories or freely associated states. Additionally, in all cases, these countries score substantially higher on Factor 2 than Factor 1, suggesting that democratic accountability and civil freedoms are commonplace despite limited institutional capacity, likely capped in size due to the small population. The exception to this phenomenon is the small island nation of Niue.

The geopolitical factors contributing to these anomalies will be further discussed in the next section.

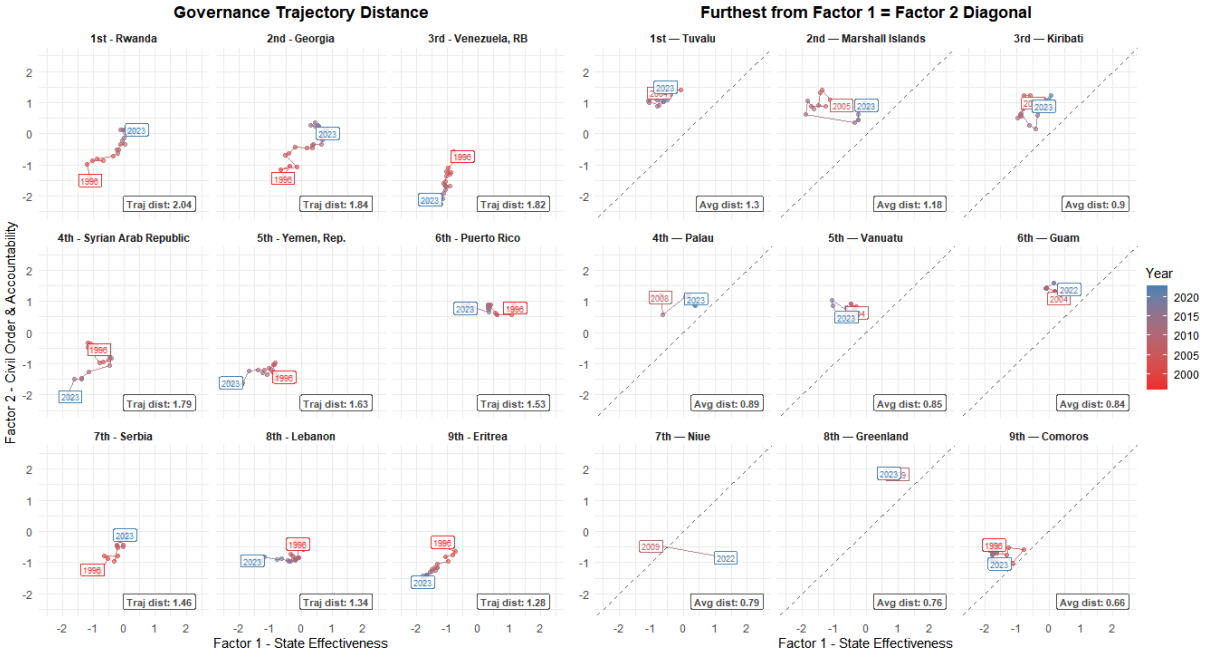


Figure 5: **Country-Level Governance Profiles.** *Left:* The nine countries with the largest Euclidean trajectory distance in factor space (shown in the bottom right of each subplot) between the early (1996–2000) and recent (2019–2023) windows, with Rwanda, Georgia, and Serbia showing the greatest improvement, and Venezuela, Syria, and Yemen the greatest decline. *Right:* Countries whose average factor scores deviate furthest from the Factor 1 = Factor 2 diagonal, led by Tuvalu, the Marshall Islands, and Kiribati, all small island nations scoring substantially higher on Civil Order & Accountability than State Effectiveness.

7 Discussion

We will first look to answer our research questions and then provide additional context around interesting phenomena found in our data.

7.1 Is Governance a Multi-Dimensional Construct?

The parallel analysis, Kaiser criterion, and cumulative variance results converge on a single dominant factor accounting for 97.41% of variance, suggesting that governance quality is largely one-dimensional. However, in retaining a second factor, meaningful nuances emerged in the separation of political stability found in Factor 1 and institutional indicators found in Factor 2 under Promax rotation.

Factor 1, hereafter will be referred to as State Effectiveness, as the strong loadings of Government Effectiveness (0.93), Regulatory Quality (0.71), and Control of Corruption (0.48), collectively may be interpreted as reflecting the government's ability to formulate, implement, and enforce public policy without being undermined by private interests to corrupt. In a similar fashion, Factor 2 will be referred to as Civil Order & Accountability, since strong loadings of Voice & Accountability (0.72), Rule of Law (0.68), Political Stability (0.72), and Control of Corruption (0.51), reflect the degree to which citizens feel safe, represented, and protected under their government.

7.2 Regional Differences in Governance

Regional differences in governance profiles are substantial across the observation period, Western Europe consistently occupies the highest position on both factors, reflecting long-established institutional frameworks and collaboration brought about by the European Union. The Middle East & North Africa region has slightly reduced Civil Order & Accountability (Factor 2) scores to State Effectiveness (Factor 1), reflecting states with functioning bureaucracies, but limited political and civil freedoms. This is a pattern of rentier states that have an over-reliance on oil and other natural resources (Ross, 2001).

The bimodal distributions observed within North America, South America, and South Asia suggest that geographic groupings alone are insufficient to capture governance similarity. These within-region differences motivate future work using clustering techniques, such as Gaussian Mixture Models, to identify governance clusters not accounted for. Some such clusters may be more inclined to group countries by culture and its influence on government.

7.3 How have Regional Differences Changed over Time?

The bootstrap analysis reveals that regional governance trajectories have differed. Eastern Europe & Central Asia showed the largest improvement ($p = 0.004$) with countries such as Georgia and Serbia being identified in country-level analysis as some of the highest governance growth (Figure 5), reflecting successful post-conflict democratic transitions.

By Contrast, Central America & Caribbean ($p = 0.018$) and Western Europe ($p = 0.014$) both regressed on Factor 1. The Western decline is a notable finding that aligns with some beliefs that large-scale migration is negatively impacting these regions of the world (Norris and Inglehart, 2020). The South American decline is consistent with the documented deterioration of democracy in countries such as Venezuela after President Chavez's departure and the continuing depletion of natural resources in nations abroad (Vera, 2015).

7.4 The Case of Small Island Nations

A recurring theme in the country-level analysis is the similarities found in the governance profiles of small island nations scoring substantially higher on Civil Order & Accountability

(Factor 2) than State Effectiveness (Factor 1). This pattern agrees with literature documenting the adoption of democratic norms from colonial or freely associated relationships (Srebrnik, 2004), crossed with limited bureaucratic capacity due to the small populations and resource constraints. Some examples are the Marshall Islands and Guam with the United States, while Greenland operates under Danish oversight.

Niue, another small island nation, is a particularly interesting outlier. Despite only appearing in the data for three years (2009, 2010, and 2022), it has one of the highest positive governance trajectories. This is likely a reflection of the economic and human development gains supported by its free association relationship with New Zealand, providing Niue with financial assistance, defence, and access to New Zealand citizenship (Adam Smith International, 2015). Niue is also anomalous due to its inverted relationship between State Effectiveness (Factor 1) and Civil Order & Accountability (Factor 2), contrasting that of other small island nations. This is likely due to the positive democratic accountability influence brought about by their partnership with New Zealand, which also scores high in both factors.

8 Limitations

Grouping countries by geographical regions is a rather arbitrary approach that does not guarantee similar government structures. Lastly, while the two chosen time windows (1996-2000 and 2019-2023) provide long-term coverage, other meaningful government shifts that occurred between these windows may be unaccounted for; a ribbon plot over time for each indicator and region was considered but omitted due to the visual blutter introduced by nine regions. Future studies may use different groupings, such as by cultural similarity or political system type, may prefer Mahalanobis over Euclidean distance to better account for the high covariance structure in a Promax factor space, and look at other notable events throughout history, such as the 2008 financial crisis.

9 Conclusion

Having applied Factor Analysis to the World Governance Indicators dataset, we found a two-dimensional construct reflecting the effectiveness of states to formulate, implement, and enforce public policy and the degree to which citizens feel safe, represented, and protected under their government. Regional differences are substantial, with Western Europe consistently leading, though bootstrap analysis confirms that some regional profiles have shifted significantly over time, including a notable decline in Western Europe itself. At a country level, our analysis identifies drivers of these regional trends and highlighted anomalous governance profiles, most notably those of small island nations, whose democratic tendencies inherited from their colonial past far exceed the institutional capacity of their small populations. Future studies may build on these findings through culturally informed groupings, clustering techniques, and a broader temporal scope.

AI Declaration

AI assisted with grammar in the report, generating the reference table for region assignments, and with coding.

References

Adam Smith International (2015). Evaluation of the Niue country programme. Technical report, New Zealand Ministry of Foreign Affairs and Trade.

Bartlett, M. S. (1951). The effect of standardization on a chi-square approximation in factor analysis. *Biometrika*, 38:337–344.

Kaiser, H. F. (1974). An index of factorial simplicity. *Psychometrika*, 39:31–36.

Kassambara, A. (2023). *ggcorrplot: Visualization of a Correlation Matrix using 'ggplot2'*. R package version 0.1.4.1.

Kaufmann, D., Kraay, A., and Mastruzzi, M. (2010). The worldwide governance indicators: Methodology and analytical issues. Policy Research Working Paper 5430, World Bank.

Korkmaz, S., Goksuluk, D., and Zararsiz, G. (2014). Mvn: An r package for assessing multivariate normality. *The R Journal*, 6(2):151–162.

Norris, P. and Inglehart, R. (2020). *Cultural Backlash: Trump, Brexit, and the Rise of Authoritarian Populism*. Cambridge University Press.

Pedersen, T. L. (2024). *patchwork: The Composer of Plots*. R package version 1.3.0.

Ross, M. L. (2001). Does oil hinder democracy? *World Politics*, 53(3):325–361.

Signorell, A. (2025). *DescTools: Tools for Descriptive Statistics*. R package version 0.99.59.

Srebrnik, H. (2004). Small island nations and democratic values. *World Development*, 32:329–341.

Vera, L. (2015). Venezuela 1999–2014: Macro-policy, oil governance and economic performance. *Comparative Economic Studies*, 57(3):539–568.

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., Takahashi, K., Vaughan, D., Wilke, C., Woo, K., and Yutani, H. (2019). Welcome to the tidyverse. *Journal of Open Source Software*, 4(43):1686.

Wilke, C. O. (2024). *cowplot: Streamlined Plot Theme and Plot Annotations for 'ggplot2'*. R package version 1.1.3.

William Revelle (2026). *psych: Procedures for Psychological, Psychometric, and Personality Research*. Northwestern University, Evanston, Illinois. R package version 2.6.1.

A Region Classification

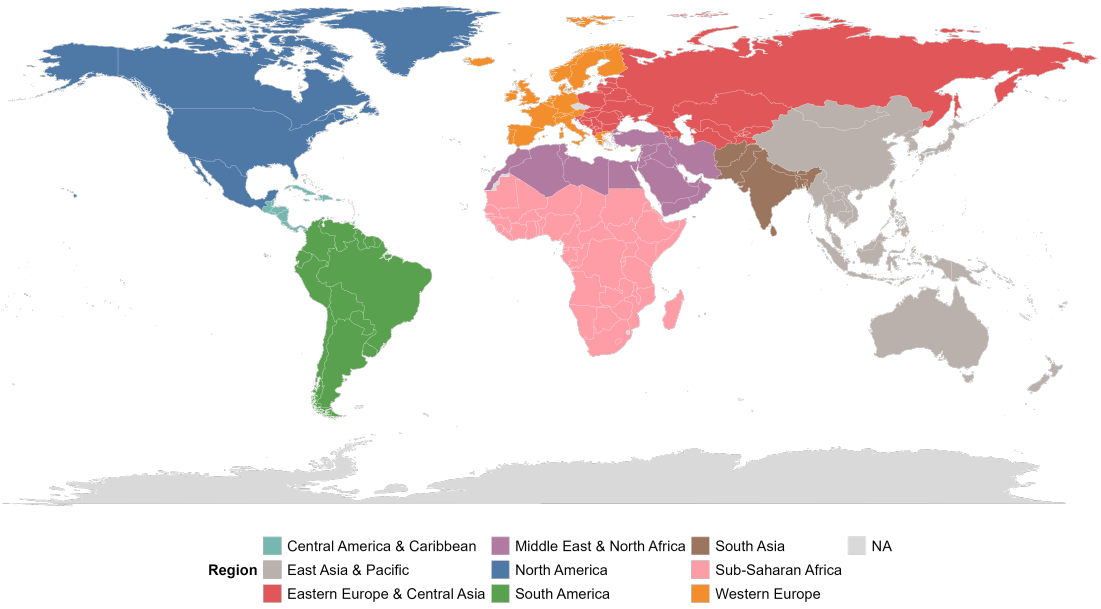


Figure 6: **Region Map Classification.** Displays the country assignments for the following regions: Sub-Saharan Africa (n = 50), East Asia & Pacific (n = 37), Eastern Europe & Central Asia (n = 29), Central America & Caribbean (n = 27), Western Europe (n = 25), Middle East & North Africa (n = 20), South America (n = 13), South Asia (n = 8), and North America (n = 5)